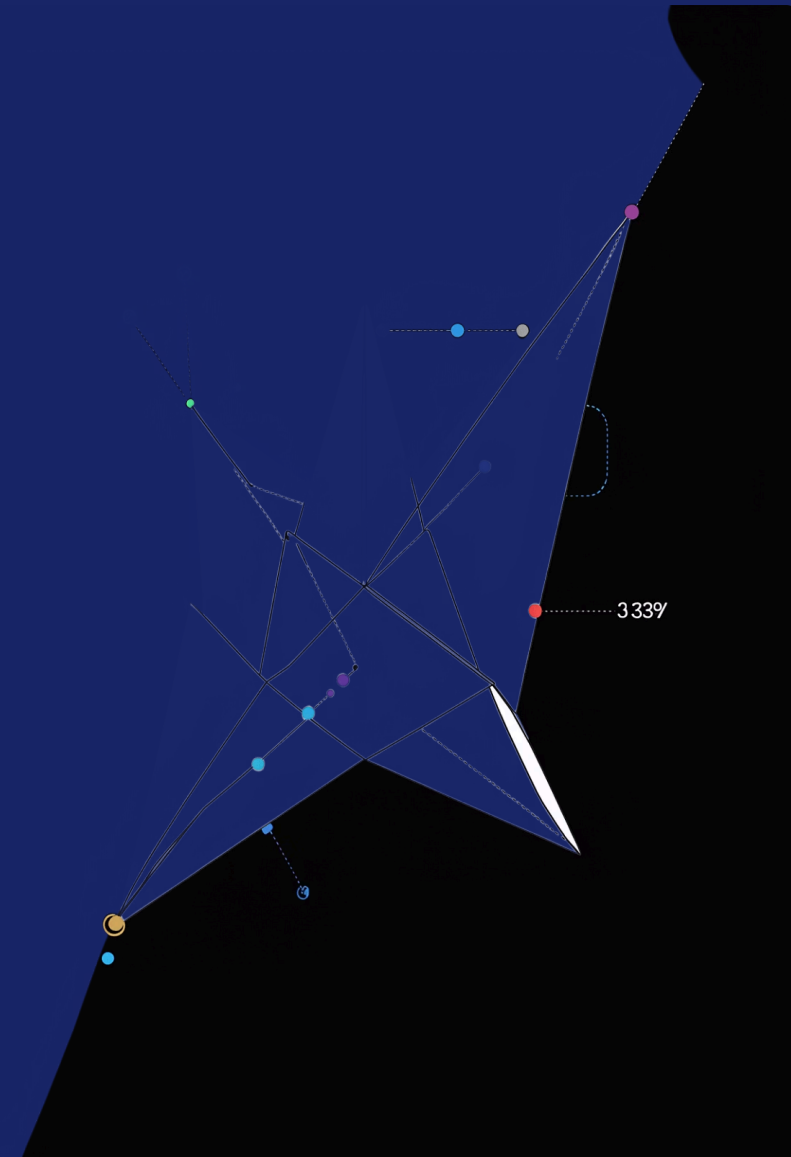


# Flat Origami is Turing Complete

How crease patterns and paper folds can simulate any computation imaginable.



# The Breakthrough (2023)

Thomas Hull and Inna Zakharevich proved that flat origami — folding paper so the finished object lies in a plane — is **Turing complete**. Their key insight: the crease lines left by folding can encode Boolean variables and logic gates.

## The Model

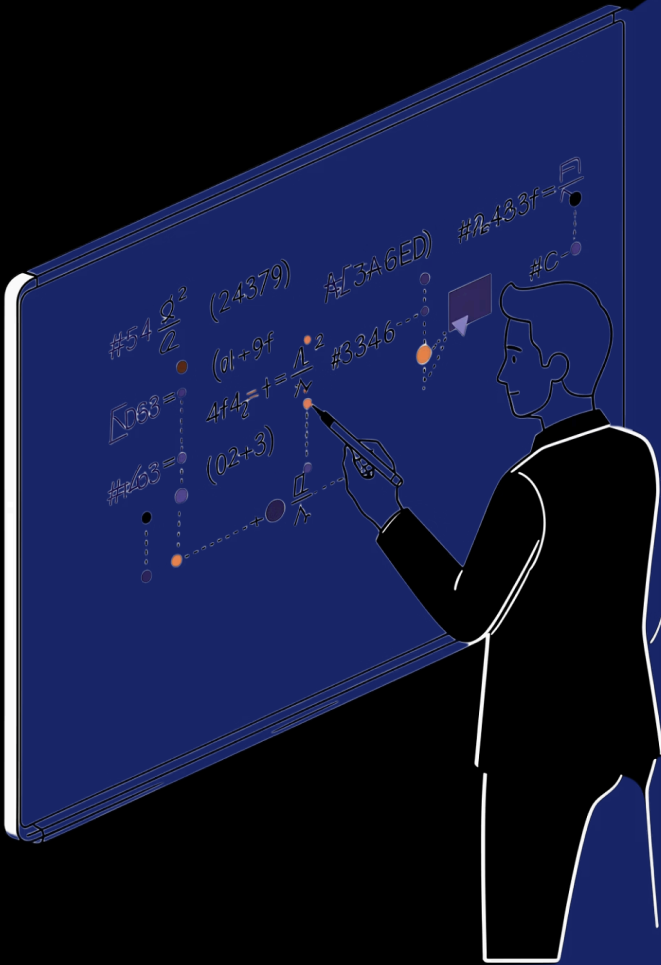
Flat origami crease patterns with **optional creases** encode logic. A crease either folds or stays flat — that's your 1 or 0.

## The Connection

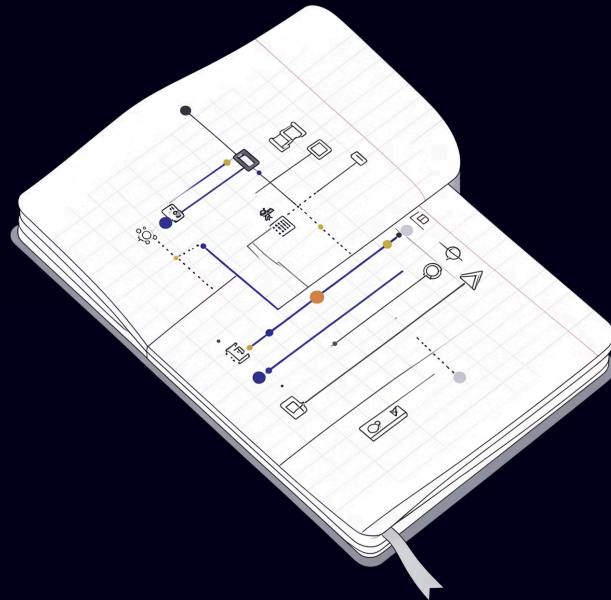
These patterns can simulate **Rule 110**, a cellular automaton proven Turing complete by Matthew Cook in 2004.

## The Implication

Any computation that can run on a computer can, in principle, be performed by **folding paper**.



# How It Works: Wires



## Boolean Signals as Fold Triplets

Information travels through the crease pattern via **wires** — triplets of parallel creases running across the paper.

- The **middle crease** is a mandatory mountain fold — always active
- The **outer two creases** are optional valley folds — only one fires at a time
- **TRUE** = the left valley fold is active
- **FALSE** = the right valley fold is active

The signal is encoded not in whether a fold exists, but in *which side* carries the active valley fold.

# The AND Gate



## Logic Through Geometry

The AND gate takes **two input wires** arriving at a central vertex at a  $2\pi/3$  angle and produces a single output wire Z.

- Output is **TRUE** only when *both* inputs are TRUE
- Built on a triangular lattice with precise crease angles
- Creases at  $\pi/3$ ,  $2\pi/3$ , and  $5\pi/6$  control the logic mechanically

❏ Kawasaki's Theorem guarantees the fold is geometrically valid: alternating sector angles at vertex O must sum to zero.

# The OR Gate

## Truth Table

| Input X | Input Y | Output Z |
|---------|---------|----------|
| FALSE   | FALSE   | FALSE    |
| FALSE   | TRUE    | TRUE     |
| TRUE    | FALSE   | TRUE     |
| TRUE    | TRUE    | TRUE     |

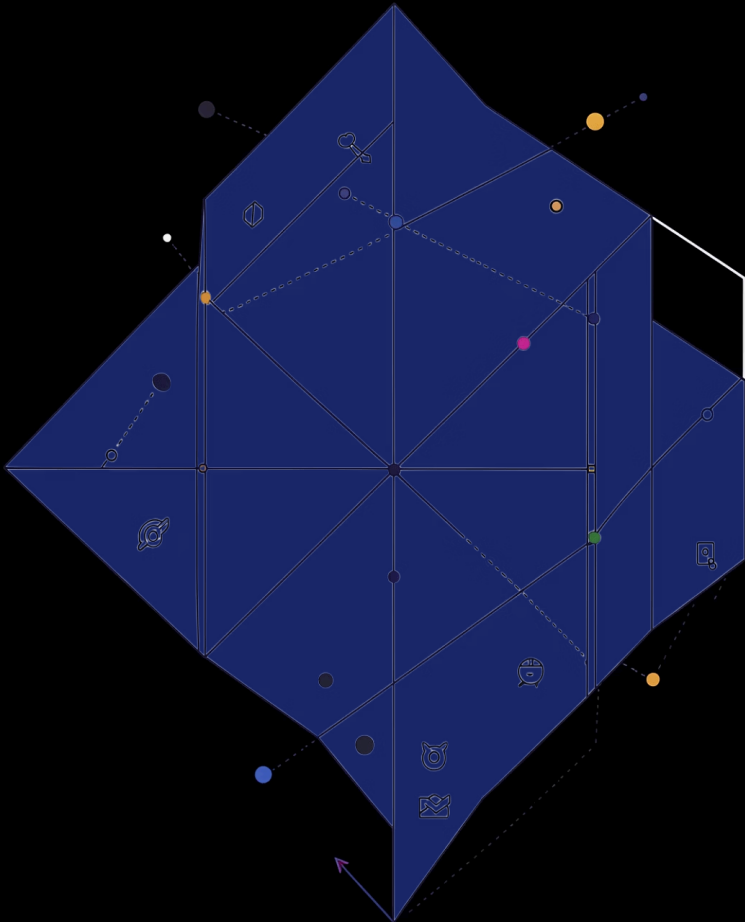
Output wire Z is TRUE whenever at least one input carries an active left valley fold.

## Mirror of AND

The OR gate uses a **similar triangular configuration** with input wires Y and X converging toward output Z — but reflected.

- Output is **FALSE** only when both inputs are FALSE
- Structurally a mirror image of the AND gate
- Crease angles force correct logic via **Kawasaki's Theorem** — the fold geometry makes any other outcome physically impossible

# The NAND Gate (Universal Gate)



## What it does

NOT-AND: output is **FALSE** only when both inputs are TRUE. In every other case, output is TRUE.

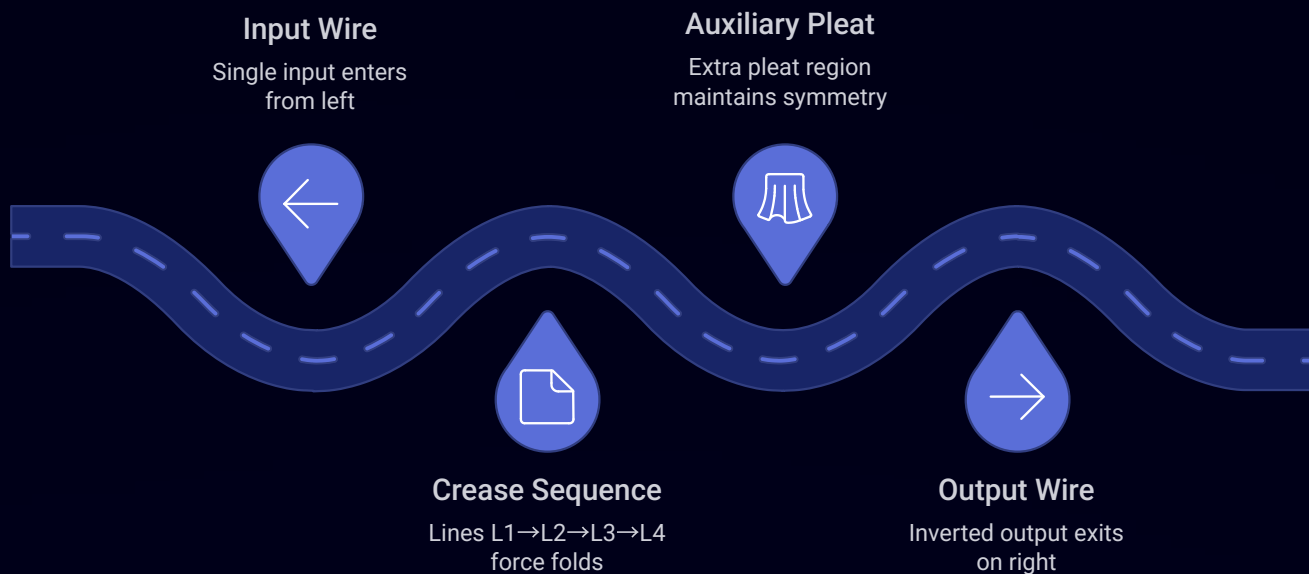
## How it's built

A reflection of the NOR gate across a vertical line. Two inputs arrive at  $2\pi/3$  angle; Proposition 6 in Hull & Zakharevich ensures exactly one valley fold per wire is ever active.

## Why it's universal

**NAND alone is sufficient** to build any logic circuit. AND, OR, NOT — every gate can be constructed from NAND gates. This makes NAND the cornerstone of Turing completeness in origami.

# The NOT Gate



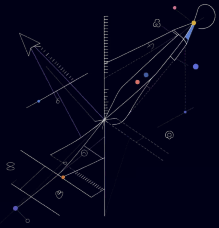
## Signal Inversion

The NOT gate is the simplest gate: it takes one input and flips it. **TRUE becomes FALSE; FALSE becomes TRUE.**

- A single input wire feeds creases  $L_1$  through  $L_4$
- Requires an **auxiliary pleat** — an extra fold region unaffected by the gate's logic
- Left-right mirror symmetry in the crease pattern forces the inversion mechanically

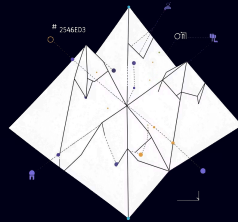
The forced fold sequence means the paper has *no physical choice* but to output the opposite signal — geometry enforces Boolean logic.

# Key Mathematical Principles



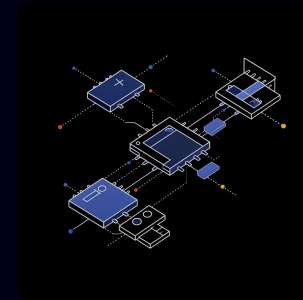
## Kawasaki's Theorem

At any interior vertex, a crease pattern folds flat **if and only if** the alternating sum of sector angles equals zero. This is what guarantees each gate vertex is physically foldable.



## Maekawa's Theorem

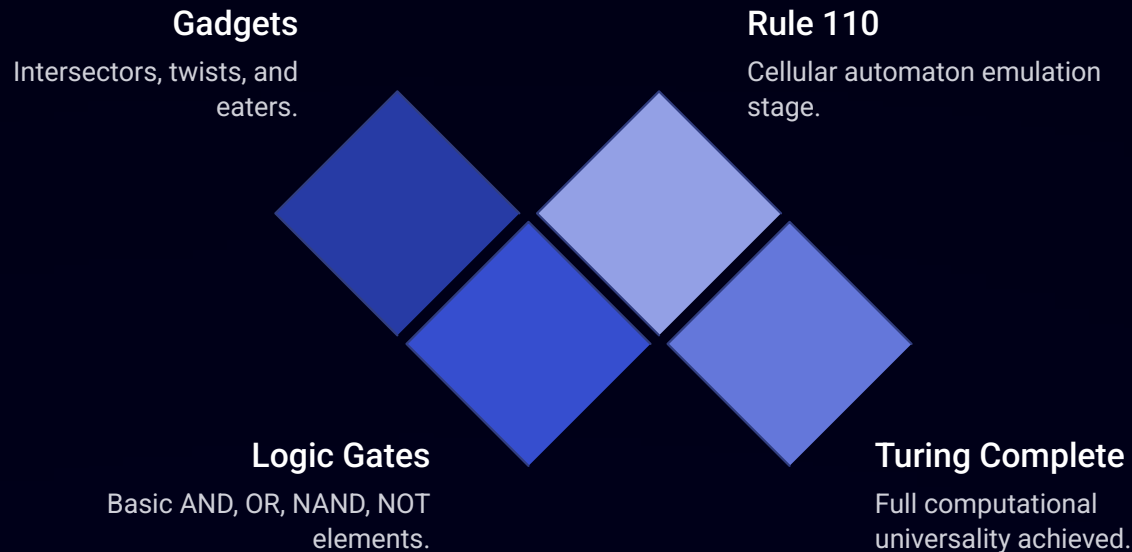
At any flat-foldable vertex, the number of mountain folds minus the number of valley folds equals  $\pm 2$ . This constrains exactly how many valley folds can be active — forcing wires to carry exactly one active signal.



## Forced Folds

Together, these theorems make alternative folding states **geometrically impossible**. The paper enforces logic — not software, not electronics. Pure constraint.

# From Gates to Rule 110



## The Chain of Proof

Individual gates are combined into larger **gadgets** — composite crease pattern modules with specific roles:

- **Intersectors** — route signals across crossing wires without interference
- **Twists** — redirect wire direction across the sheet
- **Eaters** — terminate signals cleanly at boundaries

These gadgets assemble into a crease pattern that simulates **Rule 110** — the one-dimensional cellular automaton Matthew Cook proved Turing complete in 2004. QED: origami computes.

# Why This Matters

Origami joins a remarkable club of **unexpectedly Turing-complete systems**.

## Surprising Company

Minesweeper, Magic: The Gathering, and Conway's Game of Life are all Turing complete. Flat origami now belongs on that list.

## Geometry as Computation

Computational power can emerge purely from **geometric constraints** — no circuits, no silicon, no electricity required.

## Real Applications

Opens doors to mechanical computing, programmable matter, and self-folding structures that process information physically.

## Fundamental Complexity

Paper folding is **fundamentally more complex** than anyone previously thought. The mathematics of origami touches the deepest questions in computation theory.

